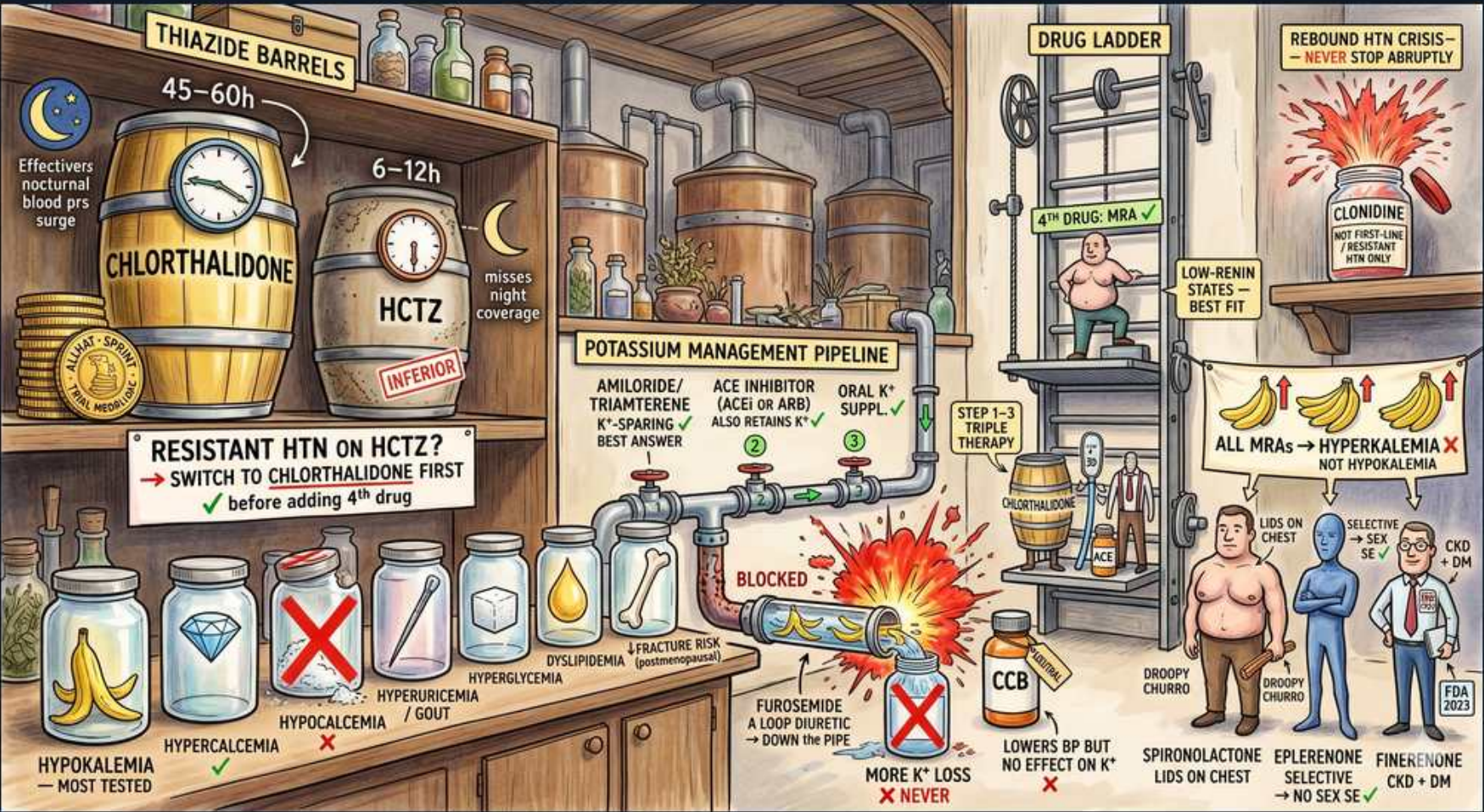


# Hypertension — Thiazides, Resistant HTN & Potassium Management





1. Chlorthalidone barrel (45-60h)

WHAT YOU SEE

Large barrel CHLORTHALIDONE, clock 45-60h, note covers nocturnal surge.

WHAT IT ENCODES

Chlorthalidone has a long half-life (~45-60 hours), giving true 24-hour BP control that covers the early-morning surge and nighttime BP — when CV events cluster. Dose 12.5-25 mg daily. It is the thiazide with the strongest outcome evidence (ALLHAT, SHEP) and is ~1.5-2x more potent than HCTZ milligram-for-milligram. Mechanism: blocks the Na-Cl cotransporter in the distal convoluted tubule.

BOARD TRAP

Picking HCTZ over chlorthalidone for a patient who needs reliable 24-hour control — WRONG when outcomes matter. The discriminator: chlorthalidone's long half-life covers nocturnal/morning BP and has better CV outcome data, so it is the preferred thiazide on the boards.

2. HCTZ barrel (6-12h, inferior)

WHAT YOU SEE

Smaller barrel HCTZ, clock 6-12h, misses night coverage, INFERIOR.

WHAT IT ENCODES

Hydrochlorothiazide is shorter-acting (half-life ~6-12 hours), so a once-daily dose does not control overnight and early-morning BP, and it has weaker CV-outcome evidence than chlorthalidone. It remains the most prescribed thiazide and is fine in fixed-dose combinations, but is considered inferior when head-to-head with chlorthalidone for hard endpoints.

BOARD TRAP

Assuming HCTZ and chlorthalidone are equivalent at the same milligram dose — WRONG. The discriminator: HCTZ is less potent, shorter-acting (misses the nocturnal surge), and has inferior outcome data; switch to chlorthalidone before declaring a thiazide has failed.

3. Resistant HTN -> chlorthalidone first

WHAT YOU SEE

Banner: RESISTANT HTN ON HCTZ? SWITCH TO CHLORTHALIDONE FIRST before adding 4th drug.

WHAT IT ENCODES

Resistant HTN = BP above goal despite three optimally dosed first-line agents INCLUDING a diuretic, or BP controlled but requiring >=4 drugs. Before adding a 4th agent, confirm adherence, rule out white-coat effect (ABPM), check for secondary causes, remove offending agents (NSAIDs, decongestants, OCPs, alcohol, stimulants), and optimize the diuretic — often by switching HCTZ to long-acting chlorthalidone.

BOARD TRAP

Adding a fourth antihypertensive to a patient on HCTZ + ACEi + amlodipine without first optimizing the diuretic or excluding pseudoresistance — WRONG. The discriminator: confirm adherence, exclude white-coat HTN/secondary causes, stop BP-raising drugs, and switch to chlorthalidone before escalating to a 4th agent.

4. Thiazide adverse effects panel

WHAT YOU SEE

Row: HYPOKALEMIA (most tested), HYPERCALCEMIA, hypocalemia X, HYPERURICEMIA/GOUT, HYPERGLYCEMIA, DYSLIPIDEMIA, fracture-risk.

WHAT IT ENCODES

Thiazide metabolic effects (mnemonic 'hyperGLUC' + low K/Na): HYPOkalemia (most tested), HYPOnatremia (can be severe, esp. elderly women), HYPERcalcemia (Ca retained — useful in osteoporosis/calcium stones), HYPERuricemia/gout, HYPERglycemia, and dyslipidemia. Sulfa allergy is a relative caution. They reduce calcium excretion, lowering kidney-stone and fracture risk.

BOARD TRAP

Saying thiazides cause hypocalcemia, or choosing a thiazide as the cause of a calcium kidney stone — WRONG. The discriminator: thiazides cause HYPERcalcemia (decrease urinary Ca) and are actually protective against calcium stones; it is LOOP diuretics that cause hypocalcemia and promote stones.

5. Potassium management pipeline

WHAT YOU SEE

PROTASSIUM MANAGEMENT header over branching pipes.

WHAT IT ENCODES

Diuretic-induced hypokalemia is managed by choosing K-retaining strategies rather than just supplementing. Options branch into: add a K-sparing diuretic (amiloride/triamterene), leverage a RAAS blocker (ACEi/ARB) which retains K, or give oral KCl. Correct magnesium too — refractory hypokalemia often won't resolve until hypomagnesemia is fixed.

BOARD TRAP

Treating persistent thiazide hypokalemia with escalating oral KCl while ignoring a low magnesium — WRONG. The discriminator: hypomagnesemia drives renal K wasting, so check and replete Mg; a K-sparing agent or ACEi/ARB is often the more durable fix than supplements alone.

6. K-sparing / ACEI-ARB retains K

WHAT YOU SEE

Branch: amiloride/triamterene K-sparing (best answer); ACEi/ARB also retains K; oral K supplement.

WHAT IT ENCODES

K-sparing diuretics amiloride and triamterene block the epithelial sodium channel (ENaC) in the collecting duct, retaining K+ to offset thiazide losses — often the best answer for diuretic-induced hypokalemia. ACE inhibitors and ARBs also raise K+ by lowering aldosterone. Oral KCl is an acceptable alternative. All of these RAISE potassium, so monitor for hyperkalemia, especially in CKD.

BOARD TRAP

Combining a K-sparing diuretic, an ACEi, AND a potassium supplement in a patient with CKD — WRONG (additive hyperkalemia and arrhythmia risk). The discriminator: pick ONE K-retaining strategy to offset thiazide loss and monitor K+; stacking several, especially in renal impairment, is dangerous.



## 7. Furosemide loop / more K loss

### WHAT YOU SEE

Exploding pipe: FUROSEMIDE = LOOP DIURETIC, MORE K LOSS, NEVER (to fix hypokalemia).

### WHAT IT ENCODES

Loop diuretics (furosemide) inhibit the Na-K-2Cl cotransporter in the thick ascending limb and cause MORE potassium and magnesium wasting plus HYPOcalcemia — they worsen, not fix, hypokalemia. Loops are reserved for HTN with significant CKD (eGFR <30, where thiazides lose efficacy) or volume overload/HFrEF, not as a routine antihypertensive or a potassium remedy.

### BOARD TRAP

Adding furosemide to correct thiazide-induced hypokalemia, or using a thiazide in a patient with eGFR <30 — both WRONG. The discriminator: loops worsen K loss (use a K-sparing agent instead), and in advanced CKD thiazides are ineffective so a loop is the correct diuretic for volume/BP.

## 8. CCB lowers BP, no K effect

### WHAT YOU SEE

Bottle CCB: LOWERS BP BUT NO EFFECT ON K.

### WHAT IT ENCODES

Calcium channel blockers lower BP through vasodilation (DHP) or AV nodal/contractility effects (non-DHP) but are potassium-NEUTRAL — they neither cause nor correct hypokalemia. This makes a CCB a useful add-on for BP control when potassium derangement is a concern, without adding to electrolyte problems.

### BOARD TRAP

Choosing a CCB to fix diuretic-induced hypokalemia — WRONG; CCBs have no effect on potassium. The discriminator: to correct hypokalemia you need a K-retaining agent (K-sparing diuretic, ACEi/ARB) or supplementation; a CCB only addresses the BP, not the K+.

## 9. Drug ladder / 4th drug MRA

### WHAT YOU SEE

Ladder with figure climbing; top rung 4TH DRUG: MRA; STEP 1-3 TRIPLE THERAPY below.

### WHAT IT ENCODES

Once a patient is on optimized triple therapy (ACEi/ARB + CCB + thiazide, ideally chlorthalidone) and still uncontrolled, the preferred 4th-line add-on for resistant HTN is a mineralocorticoid receptor antagonist — spironolactone (PATHWAY-2 trial showed it superior to bisoprolol or doxazosin as the 4th agent). Typical spironolactone dose 25-50 mg daily; monitor potassium and renal function.

### BOARD TRAP

Adding a beta-blocker or an alpha-blocker as the 4th drug for resistant HTN — WRONG per PATHWAY-2. The discriminator: spironolactone (an MRA) is the evidence-based 4th-line agent for resistant HTN, especially in the low-renin physiology typical of these patients.

## 10. Low-renin states best fit

### WHAT YOU SEE

Note: LOW-RENIN STATES = BEST FIT.

### WHAT IT ENCODES

MRAs are most effective in low-renin, salt-sensitive HTN — the physiology common in resistant HTN, primary hyperaldosteronism, Black patients, and the elderly. Resistant HTN often reflects subclinical aldosterone excess, which explains why spironolactone outperforms other 4th-line agents. Screen for primary aldosteronism (aldosterone-to-renin ratio) in resistant HTN, especially with hypokalemia.

### BOARD TRAP

Skipping a primary aldosteronism workup in resistant HTN with unprovoked hypokalemia — WRONG. The discriminator: resistant HTN + hypokalemia (or an adrenal nodule) should prompt an aldosterone-to-renin ratio; MRAs both treat the condition and signal its low-renin pathophysiology.

## 11. All MRAs -> hyperkalemia

### WHAT YOU SEE

Bananas: ALL MRAs -> HYPERKALEMIA, NOT hypokalemia.

### WHAT IT ENCODES

All mineralocorticoid receptor antagonists (spironolactone, eplerenone, finerenone) block aldosterone and therefore RAISE potassium — the hallmark adverse effect is HYPERkalemia, plus a risk of AKI. Check K+ and creatinine before starting and 1-2 weeks after initiation/up-titration; use cautiously when combined with ACEi/ARB or in CKD.

### BOARD TRAP

Adding spironolactone to a patient already on an ACEi with eGFR 35 and not rechecking potassium — WRONG (dangerous hyperkalemia). The discriminator: MRAs cause HYPERkalemia (not hypokalemia); the combination of MRA + RAAS blocker in CKD demands close potassium monitoring.

## 12. Spironolactone: gynecomastia

### WHAT YOU SEE

Droopy chest figure: SPIRONOLACTONE, LIDS ON CHEST (gynecomastia), no sex SE.

### WHAT IT ENCODES

Spironolactone is a NON-selective MRA that also antagonizes androgen receptors and has progestational activity, producing dose-dependent gynecomastia, breast tenderness, erectile dysfunction, and menstrual irregularities. These endocrine effects are the classic reason patients discontinue it. It remains first-choice MRA for resistant HTN and HFrEF when tolerated; monitor for hyperkalemia.

### BOARD TRAP

Attributing a hypertensive man's new gynecomastia to an unrelated cause while continuing spironolactone — WRONG. The discriminator: spironolactone's anti-androgen effect is the cause; switch to selective eplerenone to retain MRA benefit without the endocrine side effects.



### 13. Eplerenone: selective, no sex SE

#### WHAT YOU SEE

A male figure with a normal flat chest (next to spironolactone's droopy 'lids on chest' figure) tagged EPLERENONE — SELECTIVE — NO SEX SE — the unaffected chest shows the selective MRA spares androgen receptors, sparing gynecomastia.

#### WHAT IT ENCODES

Eplerenone is a SELECTIVE MRA — it spares androgen and progesterone receptors, so it causes far less gynecomastia and sexual dysfunction than spironolactone. It is the preferred MRA when those side effects appear or to avoid them. It is somewhat less potent and pricier, and like all MRAs causes hyperkalemia; avoid strong CYP3A4 inhibitors which raise its levels.

#### BOARD TRAP

Stopping MRA therapy entirely because spironolactone caused gynecomastia — WRONG. The discriminator: switch to selective eplerenone, which preserves the mineralocorticoid blockade and outcome benefit while eliminating the anti-androgen side effects.

### 14. Finerenone: CKD + DM

#### WHAT YOU SEE

A figure tagged FINERENONE — CKD + DM — the non-steroidal MRA whose tested niche is cardiorenal protection in diabetic kidney disease.

#### WHAT IT ENCODES

Finerenone is a NON-steroidal, selective MRA proven (FIDELIO-DKD, FIGARO-DKD) to reduce CKD progression and CV events in patients with type 2 diabetes and chronic kidney disease, on top of a maximally tolerated ACEi/ARB. It has anti-inflammatory/anti-fibrotic effects with less hyperkalemia than steroidal MRAs, though K+ monitoring is still required.

#### BOARD TRAP

Reaching for finerenone purely as an antihypertensive — WRONG framing. The discriminator: finerenone's tested, board-relevant role is cardiorenal protection in diabetic CKD added to an ACEi/ARB, not BP lowering per se; for resistant HTN the go-to MRA is still spironolactone.

### 15. Rebound HTN / never stop clonidine abruptly

#### WHAT YOU SEE

Top-right red panel: REBOUND HTN CRISIS — NEVER STOP ABRUPTLY; CLONIDINE.

#### WHAT IT ENCODES

Clonidine is a central alpha-2 agonist that lowers sympathetic outflow; it is a later-line agent for resistant HTN. Abrupt discontinuation causes a rebound hypertensive crisis (surge of catecholamines) with severe HTN, tachycardia, and headache. Always taper clonidine; for missed doses or rebound, restart clonidine (transdermal patch is an option). Beta-blockers worsen rebound via unopposed alpha stimulation.

#### BOARD TRAP

Stopping clonidine cold in a patient admitted for surgery, or giving a beta-blocker to control clonidine-withdrawal rebound — WRONG. The discriminator: clonidine must be tapered, never abruptly stopped; treat rebound by reinstating clonidine (or with a vasodilator/alpha-blocker), and avoid beta-blocker monotherapy which causes unopposed alpha vasoconstriction.